Final Project_Data Science

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Claude (Anthropic) was used to help debug R errors, explain functions and statistical concepts, and suggest approaches for structuring code and figures. I reviewed, ran, and edited all code myself and made all analytic decisions.

Load packages and data

```
# load tidyverse for data wrangling and ggplot2
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# read the covariate and gene expression data in its original state
covar <- read.csv("C:/Users/tanne/OneDrive/GradSchool/Foundations of Data
↪  Science/Project/GSE85307_covar-1.csv")
expr <- read.csv("C:/Users/tanne/OneDrive/GradSchool/Foundations of Data
↪  Science/Project/GSE85307_geneExpression.csv")
# read the probe-to-gene-name linker file
genes <- read.csv("C:/Users/tanne/OneDrive/GradSchool/Foundations of Data
↪  Science/Project/GSE85307_geneNames.csv")
```

Inspect Covariance data

```
# show variable names, types, and first values in the covariate file
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                   : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-M" ...
```

```
##  $ geoID                       : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
##  $ mat_age_yrs                 : chr  "age_years: 31.06639288" "age_years: 27.79466119" "age_years: 3
##  $ preg_preeclampsia           : chr  "pregnancy_condition: Preeclampsia" "pregnancy_condition: Norma
##  $ delivery_gestAge_wks        : chr  "gestation_weeks: 37.42857143" "gestation_weeks: 39.42857143" "
##  $ sampleCollection_gestAge_wks: chr  "blood_draw_at_gestation_week: 10" "blood_draw_at_gestation_we
##  $ batch_Microarray            : chr  "microarray_hybridization_date: 03/15/2013" "microarray_hybridi
##  $ mat_BMI                     : chr  "bmi_kg/m2: 37.52407538" "bmi_kg/m2: 20.59144273" "bmi_kg/m2: 1
##  $ mat_vitD                    : chr  "vitamind_25(0h)_ng/ml_whole_blood_baseline: 7.94" "vitamind_25
##  $ mat_asthma                  : chr  "asthma_0=healthy_1=asthmatic: 1" "asthma_0=healthy_1=asthmatic
```

```r
# show summary statistics to spot missing codes or odd values
summary(covar)
```

```
##      subjectID         geoID        mat_age_yrs  preg_preeclampsia
##  Length   :157   Length   :157   Length   :157   Length   :157
##  N.unique :157   N.unique :157   N.unique :157   N.unique :  2
##  N.blank  :  0   N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar:  9   Min.nchar: 10   Min.nchar: 13   Min.nchar: 27
##  Max.nchar:  9   Max.nchar: 10   Max.nchar: 22   Max.nchar: 33
##  delivery_gestAge_wks sampleCollection_gestAge_wks  batch_Microarray
##  Length   :157        Length   :157                 Length   :157
##  N.unique : 44        N.unique : 53                 N.unique :  8
##  N.blank  :  0        N.blank  :  0                 N.blank  :  0
##  Min.nchar: 19        Min.nchar: 32                 Min.nchar: 41
##  Max.nchar: 28        Max.nchar: 41                 Max.nchar: 41
##       mat_BMI         mat_vitD       mat_asthma
##  Length   :157   Length   :157   Length   :157
##  N.unique :140   N.unique :132   N.unique :  2
##  N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar: 20   Min.nchar: 45   Min.nchar: 31
##  Max.nchar: 22   Max.nchar: 48   Max.nchar: 31
```

Inspect gene expression data

```r
# number of rows (probes) and columns (samples)
dim(expr)
```

```
## [1] 500 158
```

```r
# first 5 column names to see how samples are labeled
names(expr)[1:5]
```

```
## [1] "X"          "GSM2264363" "GSM2264379" "GSM2264394" "GSM2264418"
```

Inspect gene name linker file

```r
# structure of the probe-to-gene linker file
str(genes)
```

```
## 'data.frame':    750 obs. of  2 variables:
##  $ PROBEID: int  8128043 7962489 7920271 7900468 8026496 7989725 8102362 7930106 8035318 8041820 ...
##  $ GENE   : chr  "CNR1" "PLEKHA8P1" "S100A4" "NFYC" ...
```

Clean covariance data

```r
# remove the label/text before each value in columns 3-10
covar[3:10] <- lapply(covar[3:10], function(x) sub(".*: ", "", x))

# convert the continuous covariates from text to numbers
covar$mat_age_yrs <- as.numeric(covar$mat_age_yrs)
covar$delivery_gestAge_wks <- as.numeric(covar$delivery_gestAge_wks)
covar$sampleCollection_gestAge_wks <- as.numeric(covar$sampleCollection_gestAge_wks)
covar$mat_BMI <- as.numeric(covar$mat_BMI)
covar$mat_vitD <- as.numeric(covar$mat_vitD)

# convert the categorical covariates to factors with readable labels
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia, levels = c("Normal",
↪  "Preeclampsia"))
covar$mat_asthma <- factor(covar$mat_asthma, levels = c(0, 1), labels = c("Healthy",
↪  "Asthma"))
covar$batch_Microarray <- factor(covar$batch_Microarray)

# confirm after cleaning
summary(covar)
```

```
##      subjectID         geoID      mat_age_yrs       preg_preeclampsia
##  Length   :157   Length   :157   Min.   :18.43   Normal      :110
##  N.unique :157   N.unique :157   1st Qu.:22.15   Preeclampsia: 47
##  N.blank  :  0   N.blank  :  0   Median :26.57
##  Min.nchar:  9   Min.nchar: 10   Mean   :26.38
##  Max.nchar:  9   Max.nchar: 10   3rd Qu.:30.14
##                                  Max.   :38.76
##
##  delivery_gestAge_wks sampleCollection_gestAge_wks   batch_Microarray
##  Min.   :26.43        Min.   :10.00                03/19/2013:45
##  1st Qu.:38.14        1st Qu.:12.14                03/15/2013:44
##  Median :39.29        Median :13.43                03/22/2013:29
##  Mean   :38.82        Mean   :14.13                03/27/2013:26
##  3rd Qu.:40.00        3rd Qu.:16.43                04/04/2013:10
##  Max.   :42.00        Max.   :18.86                03/18/2013: 1
##                                                    (Other)   : 2
##     mat_BMI         mat_vitD       mat_asthma
##  Min.   :17.95   Min.   : 5.11   Healthy:94
##  1st Qu.:22.90   1st Qu.:13.40   Asthma :63
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
```

```
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
##
```

Verify sample matching

```
# check that expression sample columns have  matching geoID in the covariate data
all(names(expr)[-1] %in% covar$geoID)
```

```
## [1] TRUE
```

```
# check that covariate geoIDs have matching expression column
all(covar$geoID %in% names(expr)[-1])
```

```
## [1] TRUE
```

```
# check how many expression probes appear in the linker file
sum(expr$X %in% genes$PROBEID)
```

```
## [1] 500
```

```
# check whether any probe IDs are repeated in the linker file
sum(duplicated(genes$PROBEID))
```

```
## [1] 0
```

Merge names with expression data and create analysis dataset

```
# attach gene names to the expression data by matching probe IDs
expr_named <- merge(genes, expr, by.x = "PROBEID", by.y = "X")

# check that many rows remain and for duplicate gene names
nrow(expr_named)
```

```
## [1] 500
```

```
sum(duplicated(expr_named$GENE))
```

```
## [1] 0
```

```
# seshape to long format: one row per gene per sample
expr_long <- pivot_longer(expr_named, cols = -c(PROBEID, GENE), names_to = "geoID",
↪  values_to = "expression")
```

```r
# join covariates onto each sample by geoID
dat <- left_join(expr_long, covar, by = "geoID")

# confirm size (78,500 rows)
dim(dat)
```

```
## [1] 78500    13
```

Define categorical variables and find candidate genes

```r
# create vitamin D status, standard deficiency cutoff of 20 ng/mL
dat$vitD_status <- factor(ifelse(dat$mat_vitD < 20, "Deficient", "Sufficient"), levels =
↪  c("Sufficient", "Deficient"))

# create obesity status,standard BMI cutoff of 30 kg/m2
dat$obese <- factor(ifelse(dat$mat_BMI >= 30, "Obese", "Not obese"), levels = c("Not
↪  obese", "Obese"))

# check which known preeclampsia-related genes are on 500-probe array
intersect(c("FLT1", "ENG", "LEP", "PGF", "VEGFA", "PAPPA2", "INHBA", "HTRA1"),
↪  expr_named$GENE)
```

```
## character(0)
```

Rank genes by pre-eclampsia difference

```r
# for each gene, compare mean expression between preeclampsia and normal pregnancies
gene_rank <- dat %>%
  group_by(GENE) %>%
  summarise(mean_diff = mean(expression[preg_preeclampsia == "Preeclampsia"]) -
  ↪  mean(expression[preg_preeclampsia == "Normal"]),
            p_value = t.test(expression ~ preg_preeclampsia)$p.value) %>%
  arrange(p_value)

# show 10 genes with strongest difference
head(gene_rank, 10)
```

```
## # A tibble: 10 x 3
##    GENE      mean_diff p_value
##    <chr>         <dbl>   <dbl>
##  1 ATG101       0.0636  0.0141
##  2 PCSK7        0.0700  0.0156
##  3 FOXI1        0.0509  0.0163
##  4 YIPF6       -0.125   0.0172
##  5 LACRT        0.0501  0.0202
##  6 SUFU         0.0528  0.0218
##  7 HNF1A-AS1    0.0549  0.0224
##  8 MOSPD1      -0.139   0.0226
```

```
##  9 RUNDC3A     -0.261   0.0279
## 10 CYBC1        0.0539  0.0285
```

```r
# show the range of expression values across genes and samples
summary(dat$expression)
```

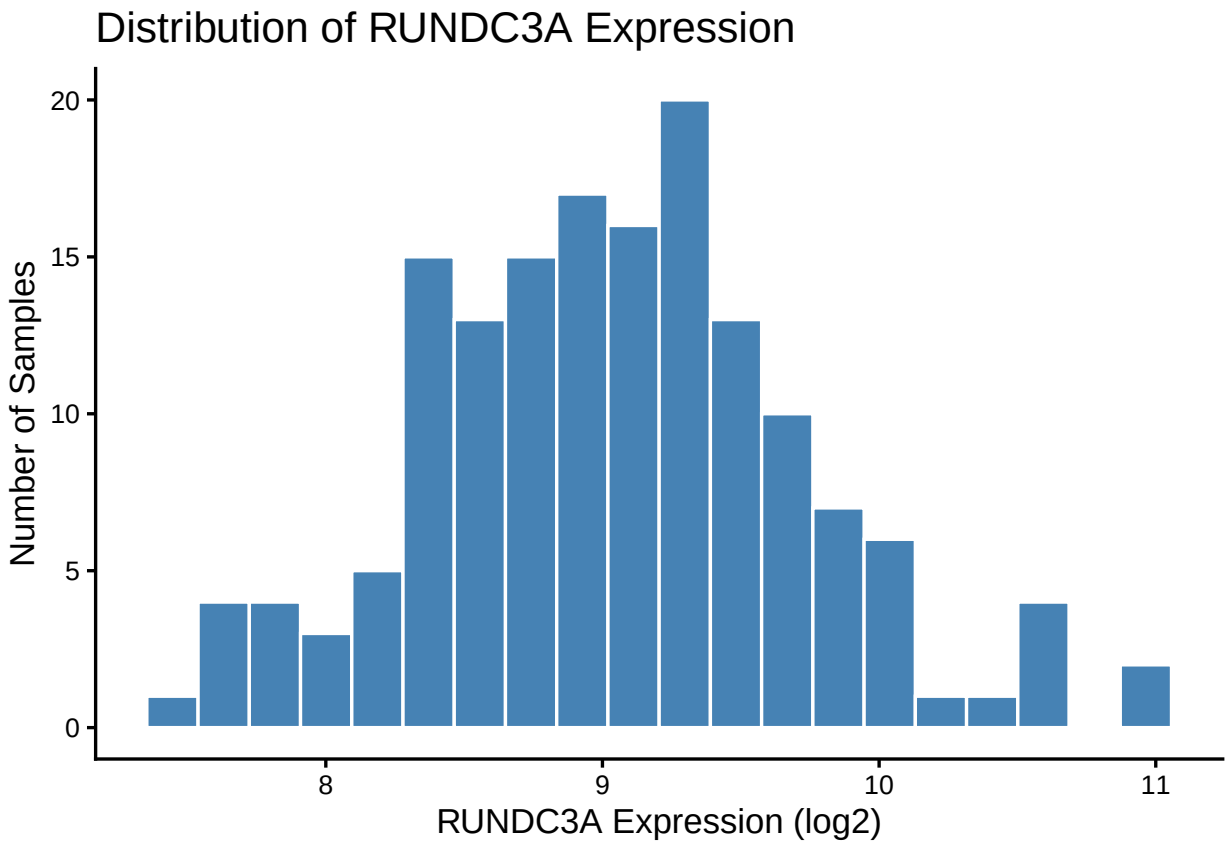
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3.124   5.725   7.047   7.123   8.290  13.215
```

Gene of interest

```r
# keep only rows for chosen gene
gene_dat <- filter(dat, GENE == "RUNDC3A")
# confirm one row per sample
nrow(gene_dat)
```

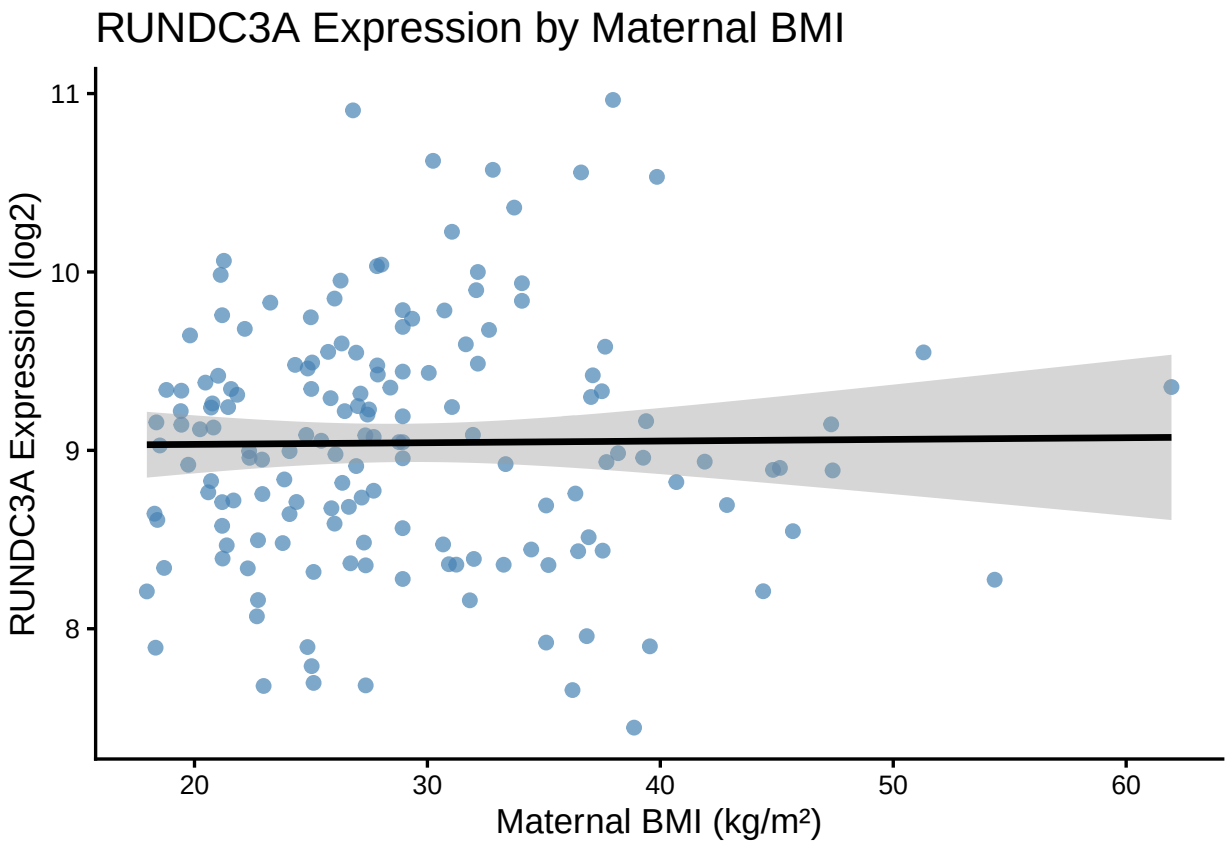
```
## [1] 157
```

```r
# plot the distribution of RUNDC3A expression across all samples
p1 <- ggplot(gene_dat, aes(x = expression)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(x = "RUNDC3A Expression (log2)", y = "Number of Samples",
       title = "Distribution of RUNDC3A Expression") +
  theme_classic(base_size = 13)
p1
```



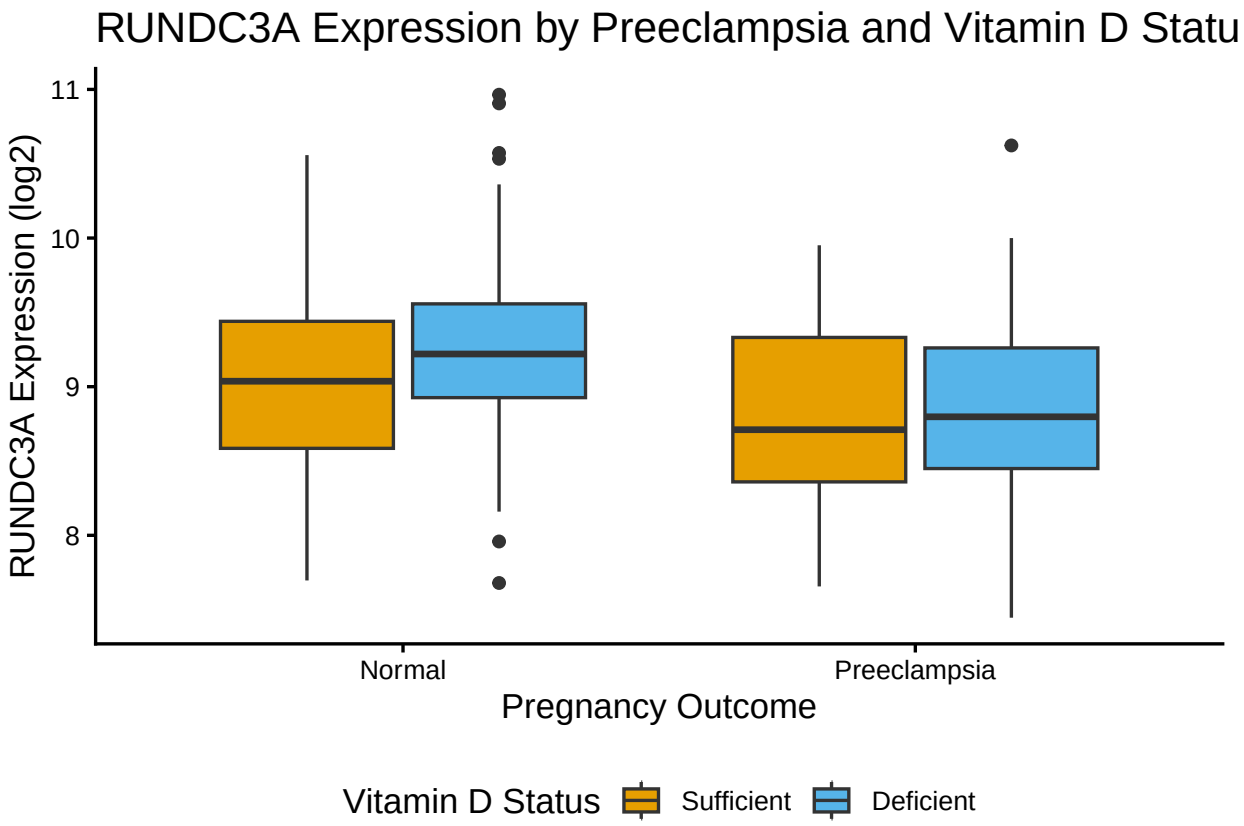
BMI vs Gene Expression

```
# Plot RUNDC3A expression against maternal BMI with a fitted regression line
p2 <- ggplot(gene_dat, aes(x = mat_BMI, y = expression)) +
  geom_point(color = "steelblue", alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black") +
  labs(x = "Maternal BMI (kg/m²)", y = "RUNDC3A Expression (log2)",
       title = "RUNDC3A Expression by Maternal BMI") +
  theme_classic(base_size = 13)
p2
```



Preeclampsia and Vit D status

```
# plot RUNDC3A expression by preeclampsia status, split by vitamin D status
p3 <- ggplot(gene_dat, aes(x = preg_preeclampsia, y = expression, fill = vitD_status)) +
  geom_boxplot() +
  scale_fill_manual(values = c("Sufficient" = "#E69F00", "Deficient" = "#56B4E9")) +
  labs(x = "Pregnancy Outcome", y = "RUNDC3A Expression (log2)",
       fill = "Vitamin D Status", title = "RUNDC3A Expression by Preeclampsia and Vitamin
       ↪  D Status") +
  theme_classic(base_size = 13) +
  theme(legend.position = "bottom")
p3
```



Save figures

```
# Save each plot as png
ggsave("fig1_histogram.png", p1, width = 7, height = 5)
ggsave("fig2_scatter.png", p2, width = 7, height = 5)
ggsave("fig3_boxplot.png", p3, width = 7, height = 5)
```

Submission 2:

Choose genes

```
# RUNDC3A plus the next two from gene_rank
myGenes <- c("RUNDC3A", "MOSPD1", "YIPF6")

# Check that all 3 genes are in data
myGenes %in% dat$GENE
```

```
## [1] TRUE TRUE TRUE
```

Plot labels

```
# labels for variable names
myLabels <- list('mat_BMI' = bquote('Maternal BMI (kg/m'^2*')'),
                 'preg_preeclampsia' = 'Pregnancy Outcome',
                 'vitD_status' = 'Vitamin D Status')
myLabels
```

```
## $mat_BMI
## "Maternal BMI (kg/m"^2 * ")"
##
## $preg_preeclampsia
## [1] "Pregnancy Outcome"
##
## $vitD_status
## [1] "Vitamin D Status"
```

Plotting function

```
# function to make plots for each gene
plotGenes <- function(data,genelist,contVar,catVars) {
  # set seed
  set.seed(321)
  # loop through all genes
  for(i in genelist) {
    # subset data to this gene
    geneData <- data[data$GENE == i, ]
    # gene label correction after feedback
    geneLabel <- bquote(italic(.(i)) ~ 'Expression (log2)')
    # hist of gene expression
    myHist <- ggplot(geneData, aes(x = expression)) +
      geom_histogram(bins = 20, fill = 'steelblue',color = 'white') +
      labs(x = geneLabel, y = 'Number of Samples') +
      theme_classic(base_size = 16)

    # scatterplot of continuous covariate vs gene expression
    myScatter <- ggplot(geneData, aes(x = .data[[contVar]], y = expression)) +
      geom_point(color = 'steelblue', alpha = 0.7, size = 2) +
      geom_smooth(method = 'lm', formula = y ~ x, color = 'black') +
      labs(x = myLabels[[contVar]], y = geneLabel) +
      theme_classic(base_size = 16)

    #boxplot by both categorical covariates
    myBox <- ggplot(geneData, aes(x = .data[[catVars[1]]], y = expression, fill =
↪   .data[[catVars[2]]])) +
      geom_boxplot(outlier.shape = NA) +
      geom_point(position = position_jitterdodge(jitter.width = 0.2), alpha = 0.5, size =
      ↪  1.5) +
      scale_fill_manual(values = c('#E69F00','#56B4E9')) +
      labs(x = myLabels[[catVars[1]]], y = geneLabel,
           fill = myLabels[[catVars[2]]]) +
```

```
      theme_classic(base_size = 16) +
      theme(legend.position = 'bottom')

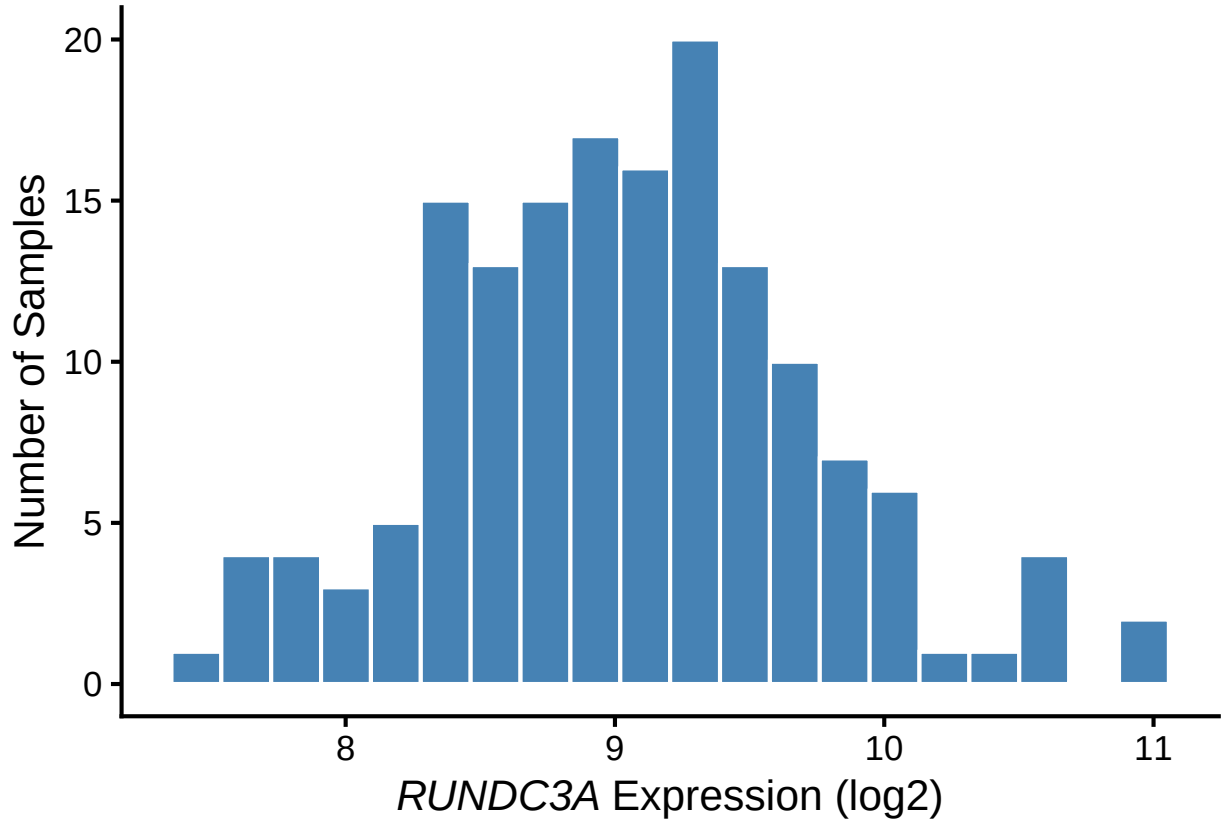
    # print 3 plots
    print(myHist)
    print(myScatter)
    print(myBox)

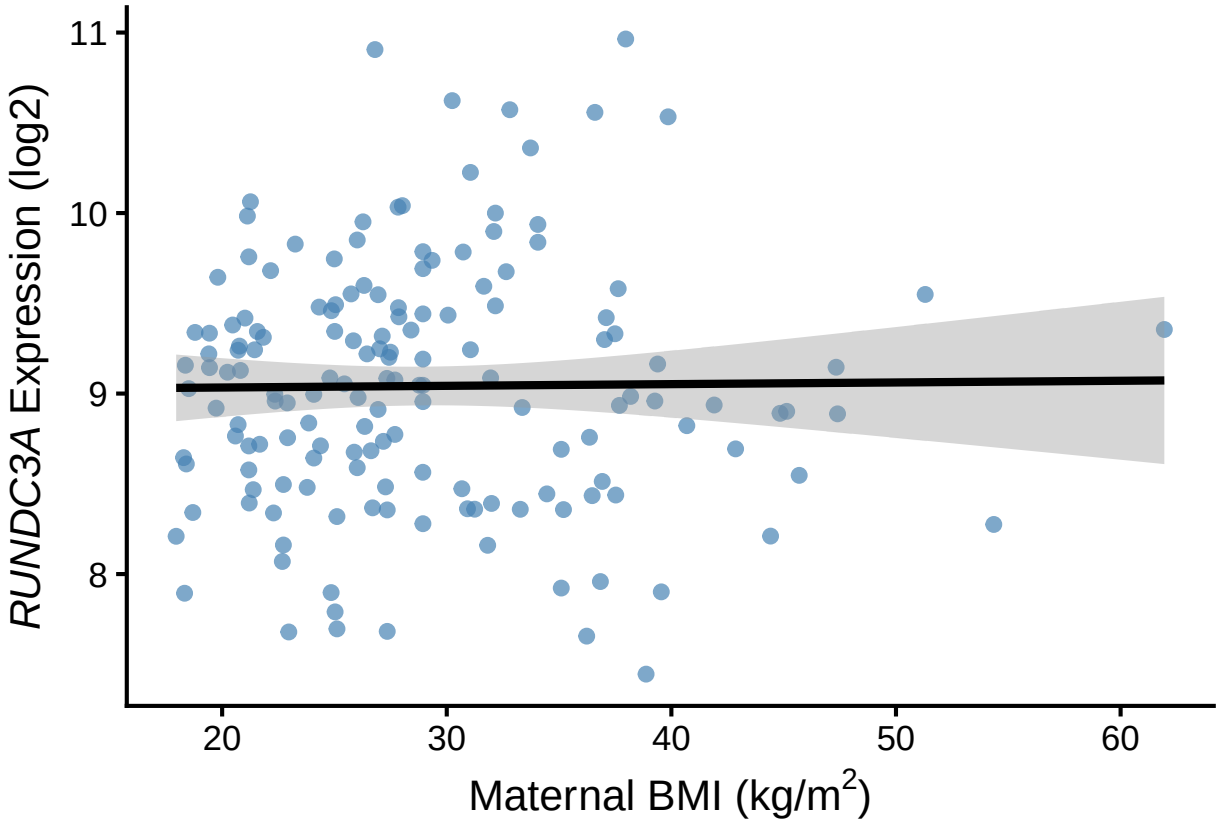
    # save 3 plots as png. named by gene
    ggsave(paste0(i, '_histogram.png'), myHist, width = 7, height = 5)
    ggsave(paste0(i, '_scatter.png'), myScatter, width = 7, height = 5)
    ggsave(paste0(i, '_boxplot.png'), myBox, width = 7, height = 5)

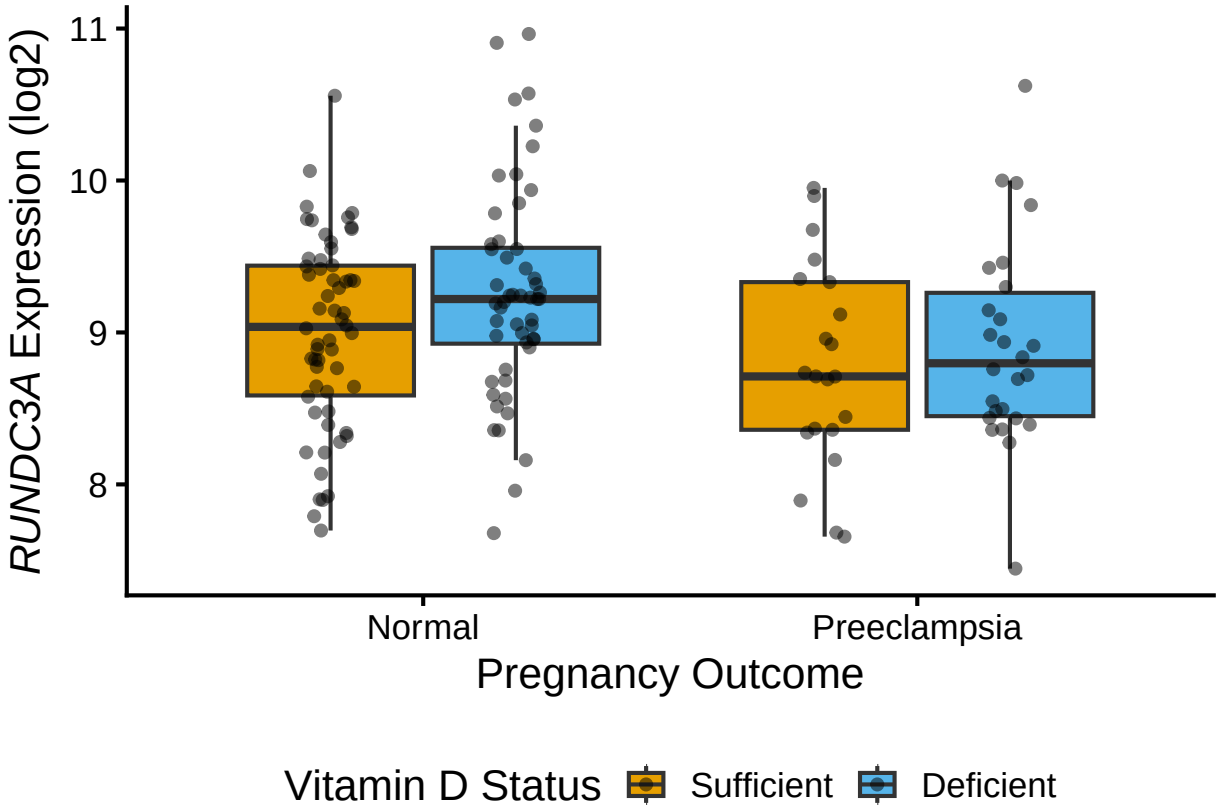
  }
}
```

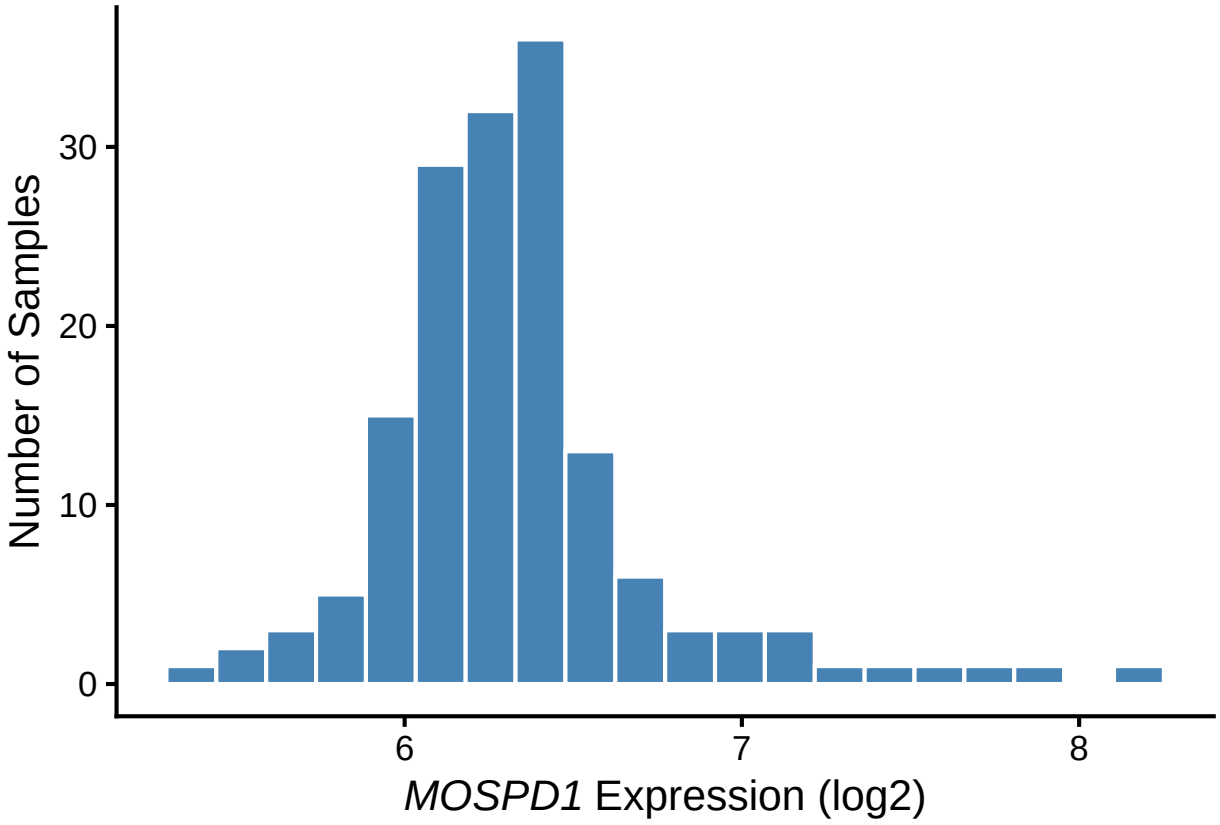
Run the function

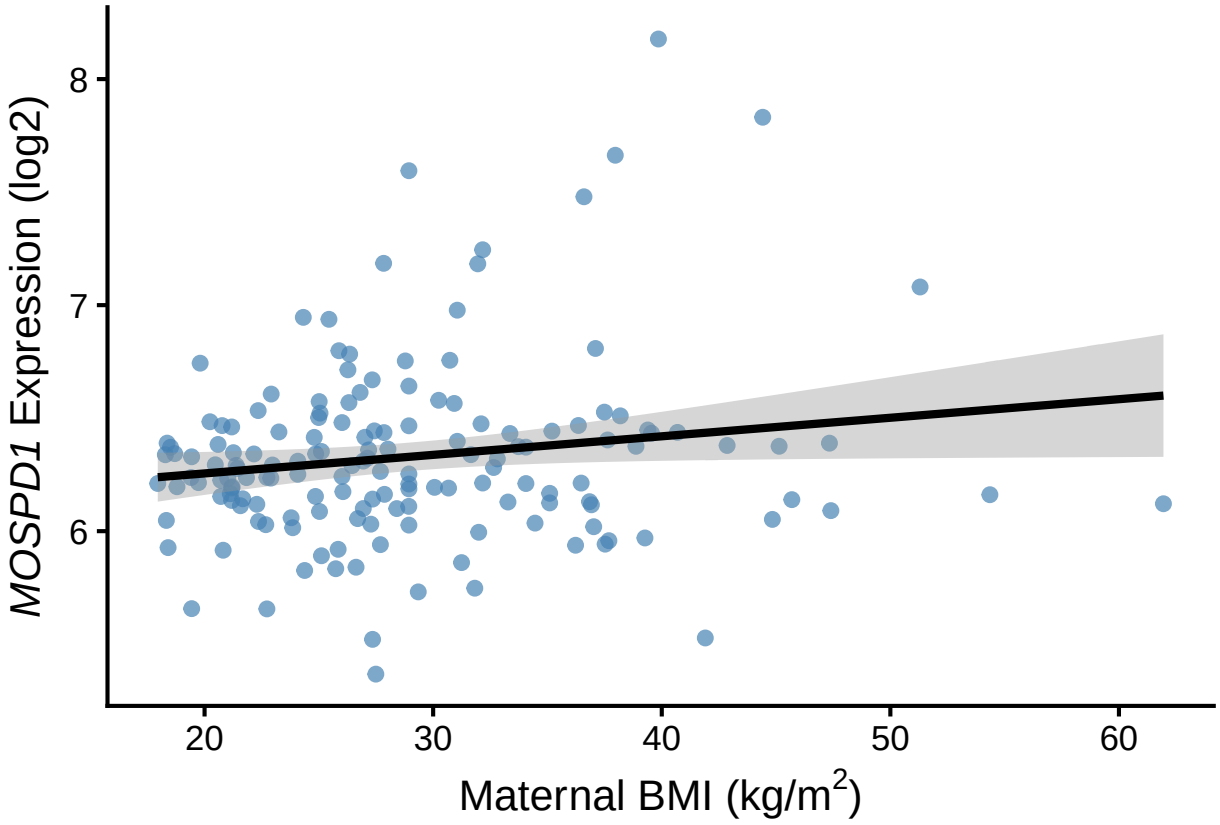
```
# 3 plot types of for each of 3 genes
plotGenes(data = dat,
          genelist = myGenes,
          contVar = 'mat_BMI',
          catVars = c('preg_preeclampsia', 'vitD_status'))
```

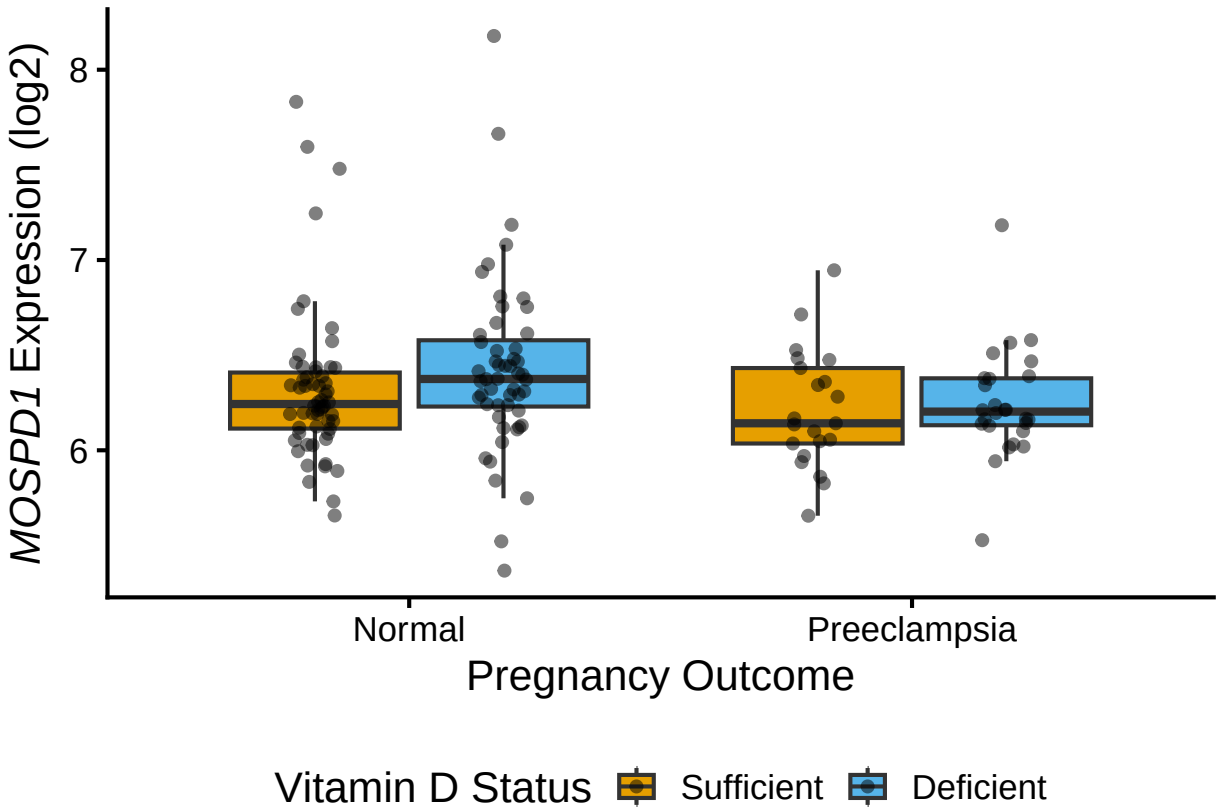


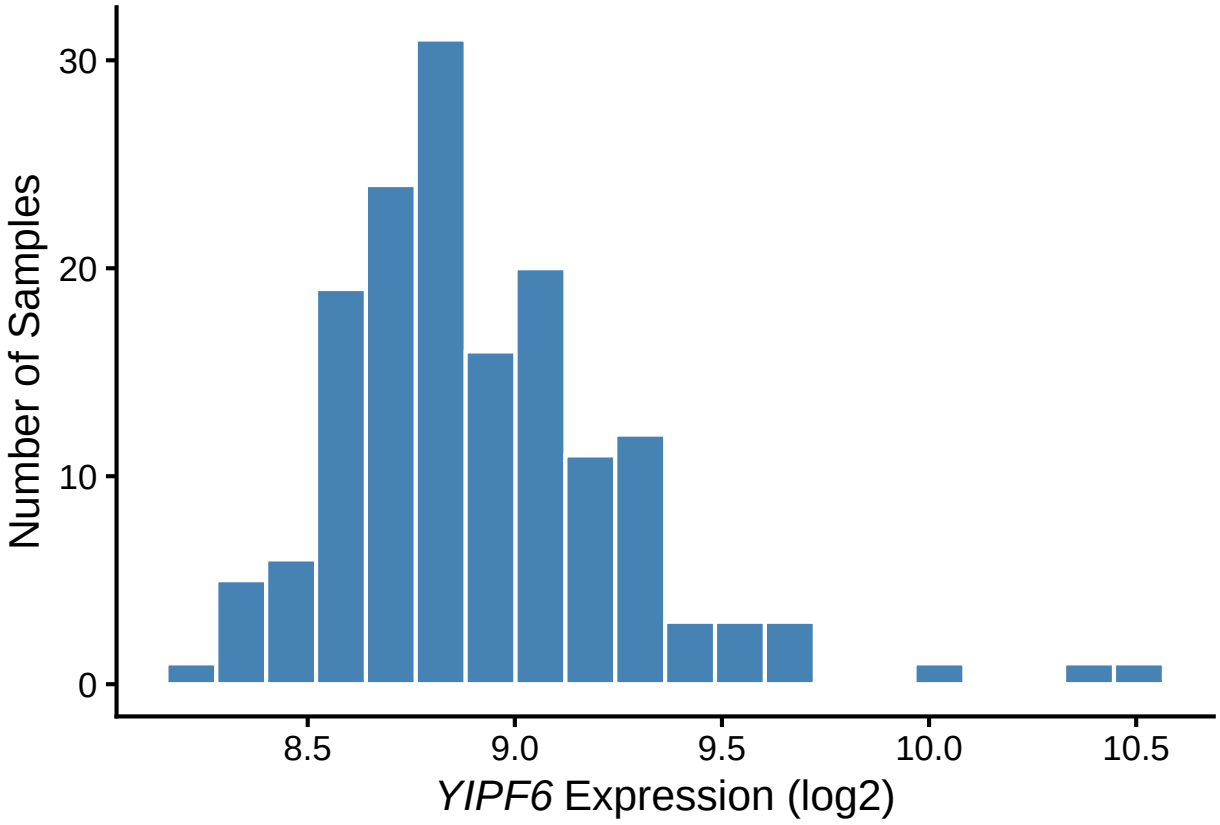

11
10
9
8
RUNDC3A Expression (log2)
20
30
40
50
60
Maternal BMI (kg/m2)

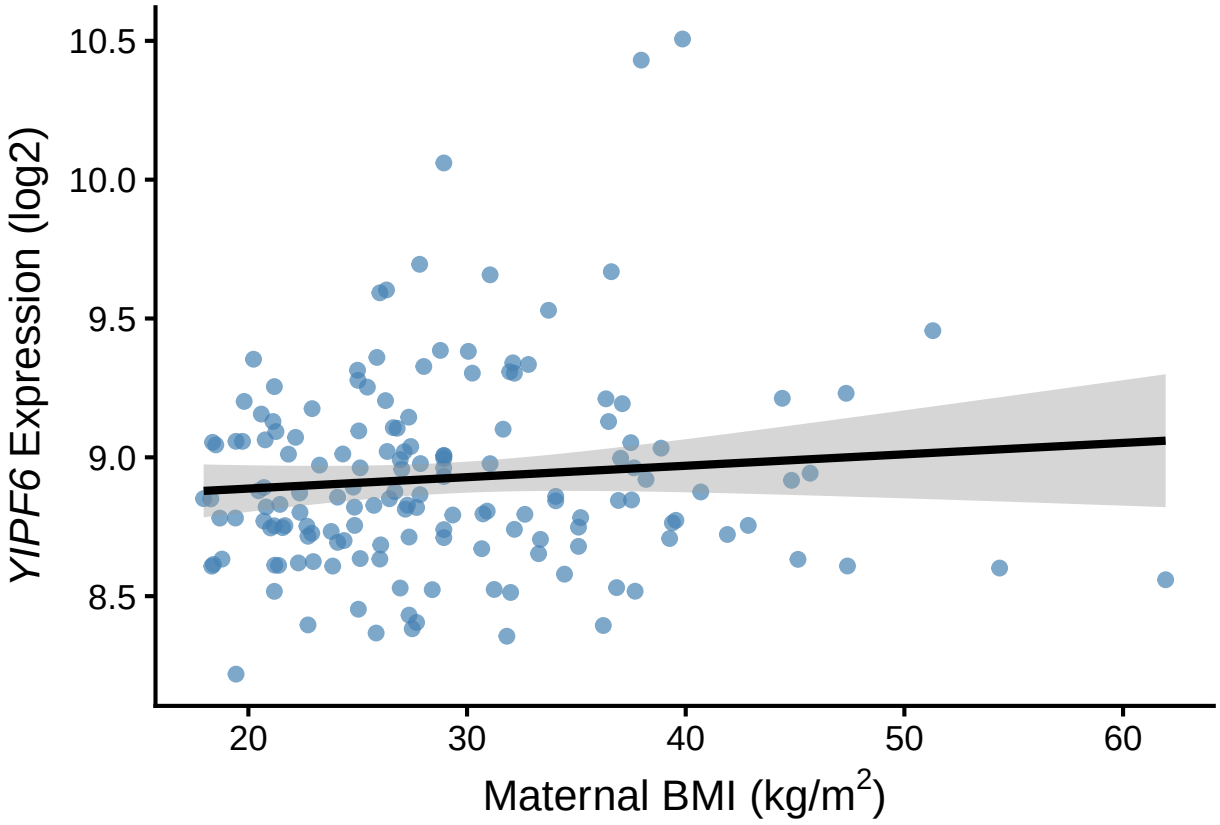

RUNDC3A Expression (log2)
11
10
9
8
Normal
Preeclampsia
Pregnancy Outcome
Vitamin D Status
Sufficient
Deficient

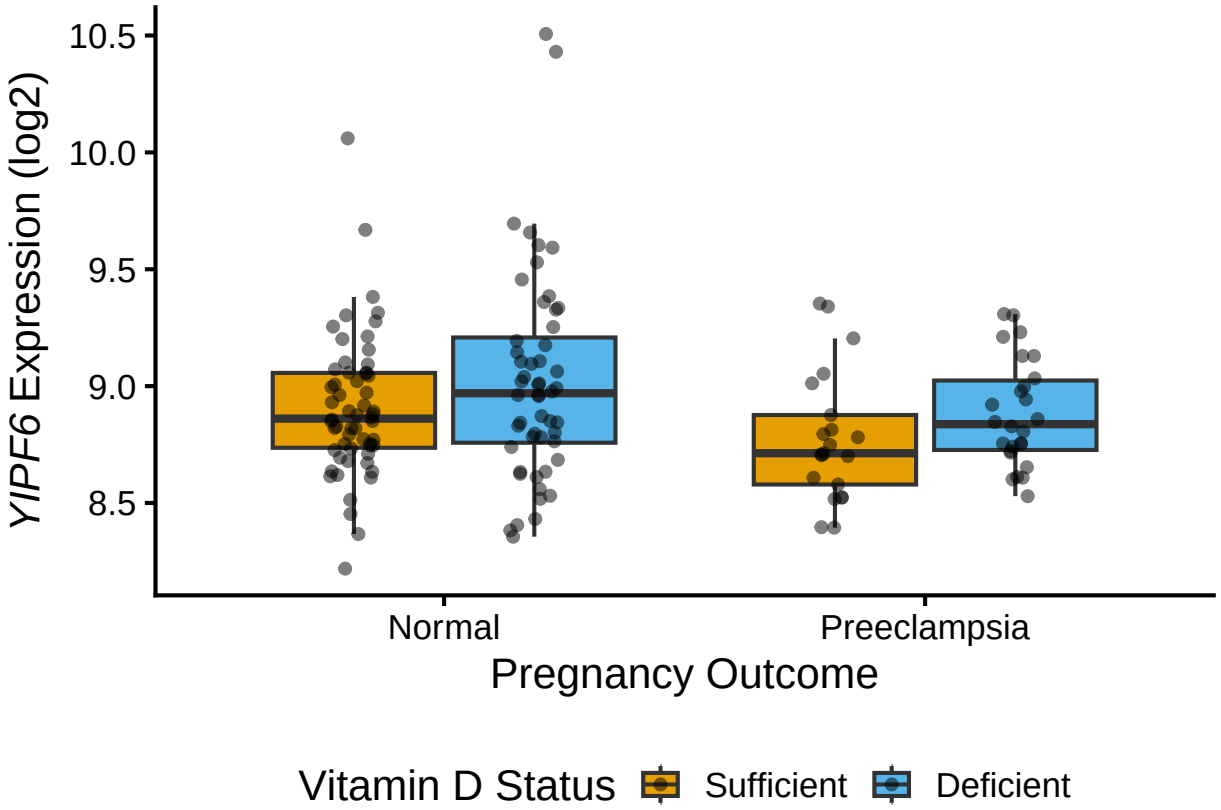

Number of Samples
30
20
10
0
6
7
8
MOSPD1 Expression (log2)


MOSPD1 Expression (log2)
8
7
6
20
30
40
50
60
Maternal BMI (kg/m2)


MOSPD1 Expression (log2)
8
7
6
Normal
Preeclampsia
Pregnancy Outcome
Vitamin D Status
Sufficient
Deficient


Number of Samples
30
20
10
0
8.5
9.0
9.5
10.0
10.5
YIPF6 Expression (log2)


10.5
10.0
9.5
9.0
8.5
YIPF6 Expression (log2)
20
30
40
50
60
Maternal BMI (kg/m2)



Final Submission start

Table One function; Add vitamin D status to the covariate data

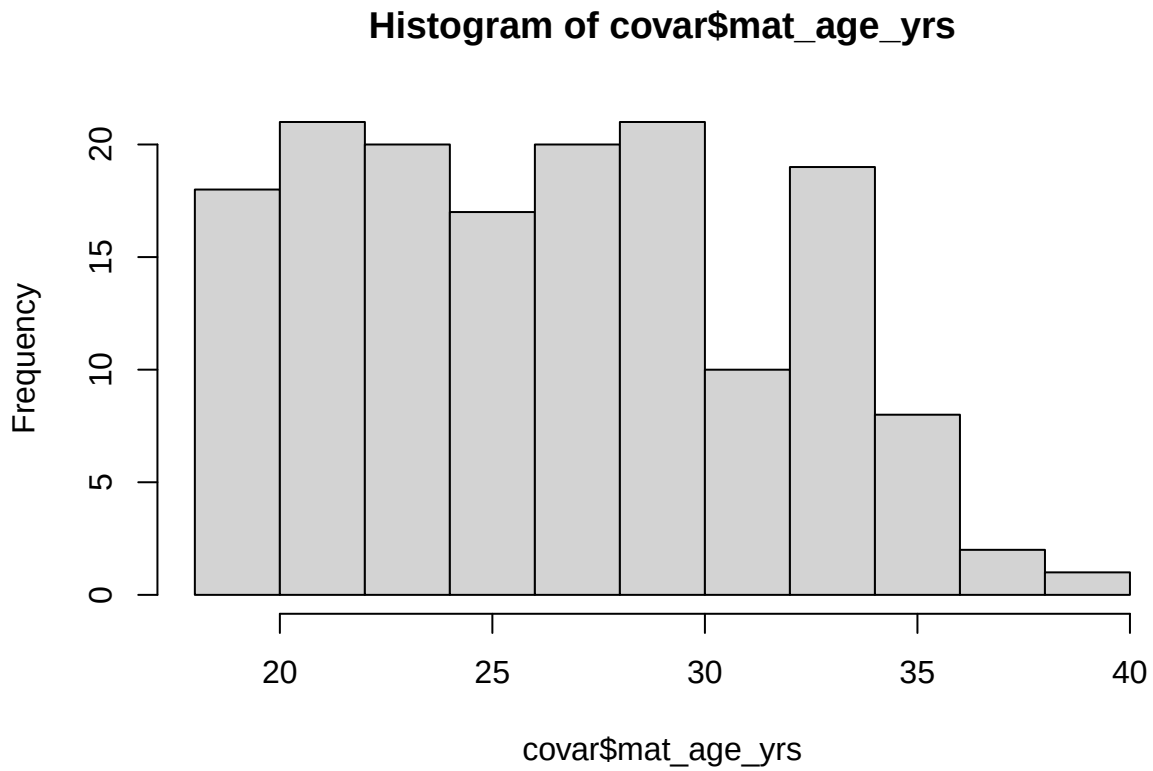
```
# function to build a table one, could not locate my previously built one
buildTableOne <- function(data,varList,nonnormVars,catVars) {

  # define list of things to loop through
  myVarList <- varList
  # define somewhere to store results
  myResults <- data.frame(row.names = myVarList,
                          'Value' = rep(NA,length(myVarList)))

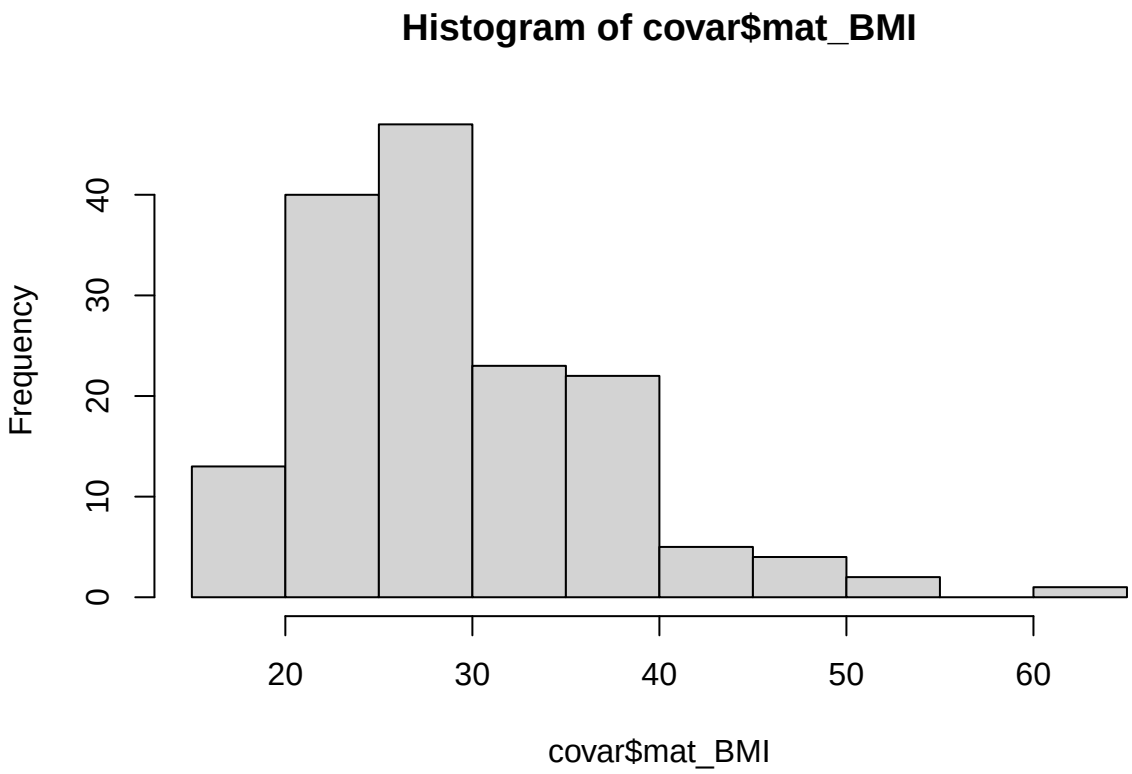
# Loop through all variables
for (i in myVarList) {
  # subset data
  myData <- data[,i]
  # categorical variables: n (%)
  if (i %in% catVars) {
    tempTab <- table(myData)
    myResults[i, 'Value'] <- paste0(names(tempTab),': ',tempTab,' (',
                                    round(tempTab/length(myData)*100,digits = 1),')',
                                    collapse = ', ')
  }
```

```
  # non-normal variables: median [IQR]
  else if (i %in% nonnormVars) {
    myResults[i, 'Value'] <- paste0(round(median(myData),digits = 2),' [',
                                    round(quantile(myData,1/4),digits = 2),', ',
                                    round(quantile(myData,3/4),digits = 2),']')
  }
  # normal variables: mean (sd)
  else {
    myResults[i, 'Value'] <- paste0(round(mean(myData),digits = 2),' (',
                                    round(sd(myData),digits = 2),')')
  }
 }
  # return results
  return(myResults)
}

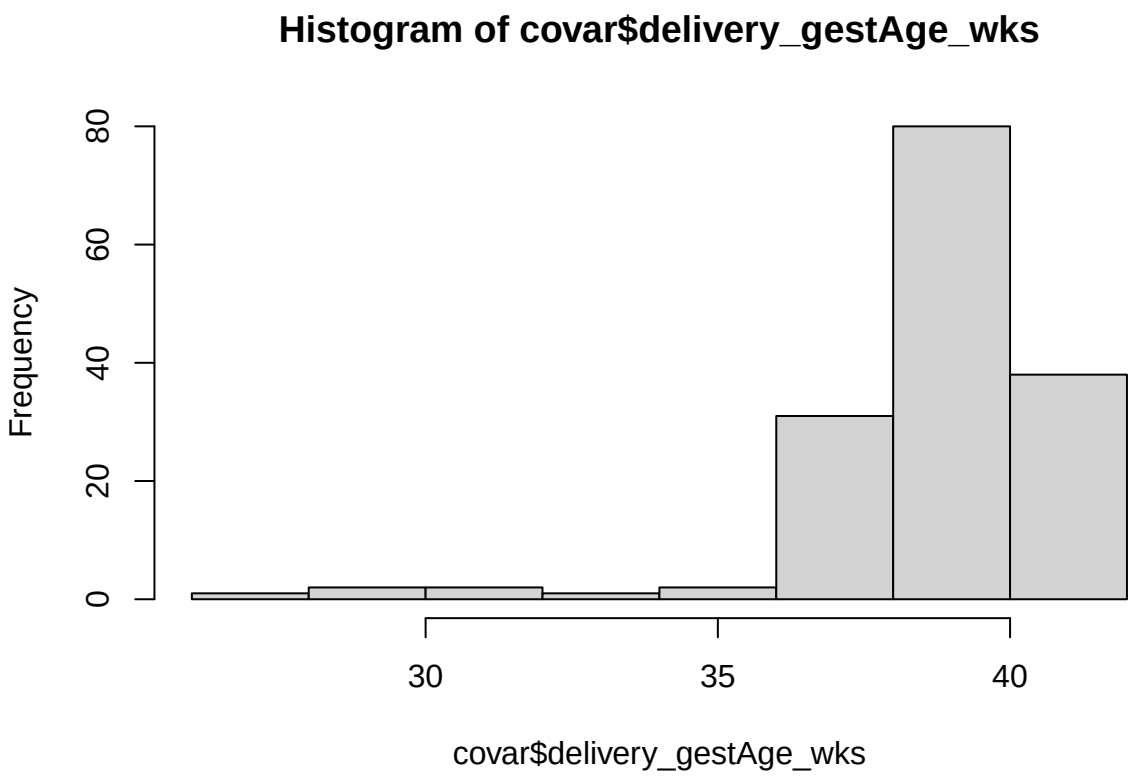
# add vitamin D status to one row per woman covariate data
covar$vitD_status <- factor(ifelse(covar$mat_vitD < 20, 'Deficient', 'Sufficient'),
                            levels = c('Sufficient', 'Deficient'))
# check distributions of variables to choose mean (SD) vs median [IQR]
hist(covar$mat_age_yrs)
```




```
hist(covar$mat_BMI)
```




```
hist(covar$delivery_gestAge_wks)
```



Build the stratified table

```
# variables to include in the table
myVars <- c('mat_age_yrs', 'mat_BMI', 'delivery_gestAge_wks', 'vitD_status')
# build column for each group and one for everyone
normalTab <- buildTableOne(data = covar[covar$preg_preeclampsia == 'Normal', ], varList =
↪  myVars, nonnormVars = c('mat_BMI', 'delivery_gestAge_wks'), catVars = 'vitD_status')
peTab <- buildTableOne(data = covar[covar$preg_preeclampsia == 'Preeclampsia', ], varList
↪  = myVars, nonnormVars = c('mat_BMI', 'delivery_gestAge_wks'), catVars =
↪  'vitD_status')
allTab <- buildTableOne(data = covar, varList = myVars, nonnormVars = c('mat_BMI',
↪  'delivery_gestAge_wks'), catVars = 'vitD_status')

# combine three columns into one table with readable row names
tableOne <- data.frame(Normal = normalTab$Value, Preeclampsia = peTab$Value, Overall =
↪  allTab$Value,
                       row.names = c('Maternal age (years), mean (SD)',
                       'Maternal BMI (kg/m$^2$), median [IQR]',
                       'Gestational age at delivery (weeks), median [IQR]',
                       'Vitamin D status, n (\\%)'))
tableOne
```

```
##                                                                    Normal
```

```
## Maternal age (years), mean (SD)                                    26.63 (5.08)
## Maternal BMI (kg/m$^2$), median [IQR]                           26.54 [22.35, 30.97]
## Gestational age at delivery (weeks), median [IQR]                  39.29 [38.57, 40]
## Vitamin D status, n (\\%)                  Sufficient: 58 (52.7), Deficient: 52 (47.3)
##                                                                          Preeclampsia
## Maternal age (years), mean (SD)                                    25.78 (4.95)
## Maternal BMI (kg/m$^2$), median [IQR]                           31.24 [24.62, 36.41]
## Gestational age at delivery (weeks), median [IQR]               39.14 [36.93, 40.07]
## Vitamin D status, n (\\%)                  Sufficient: 21 (44.7), Deficient: 26 (55.3)
##                                                                             Overall
## Maternal age (years), mean (SD)                                    26.38 (5.04)
## Maternal BMI (kg/m$^2$), median [IQR]                           27.33 [22.9, 33.27]
## Gestational age at delivery (weeks), median [IQR]                  39.29 [38.14, 40]
## Vitamin D status, n (\\%)                  Sufficient: 79 (50.3), Deficient: 78 (49.7)
```

Export to LaTeX

```
# load knitr for kable function (ran in console)
library(knitr)
# print table as latex code to paste into overleaf
kable(tableOne, format = 'latex', booktabs = TRUE, escape = FALSE,
      col.names = c('Normal (n = 110)', 'Preeclampsia (n = 47)', 'Overall (n = 157)'))
```

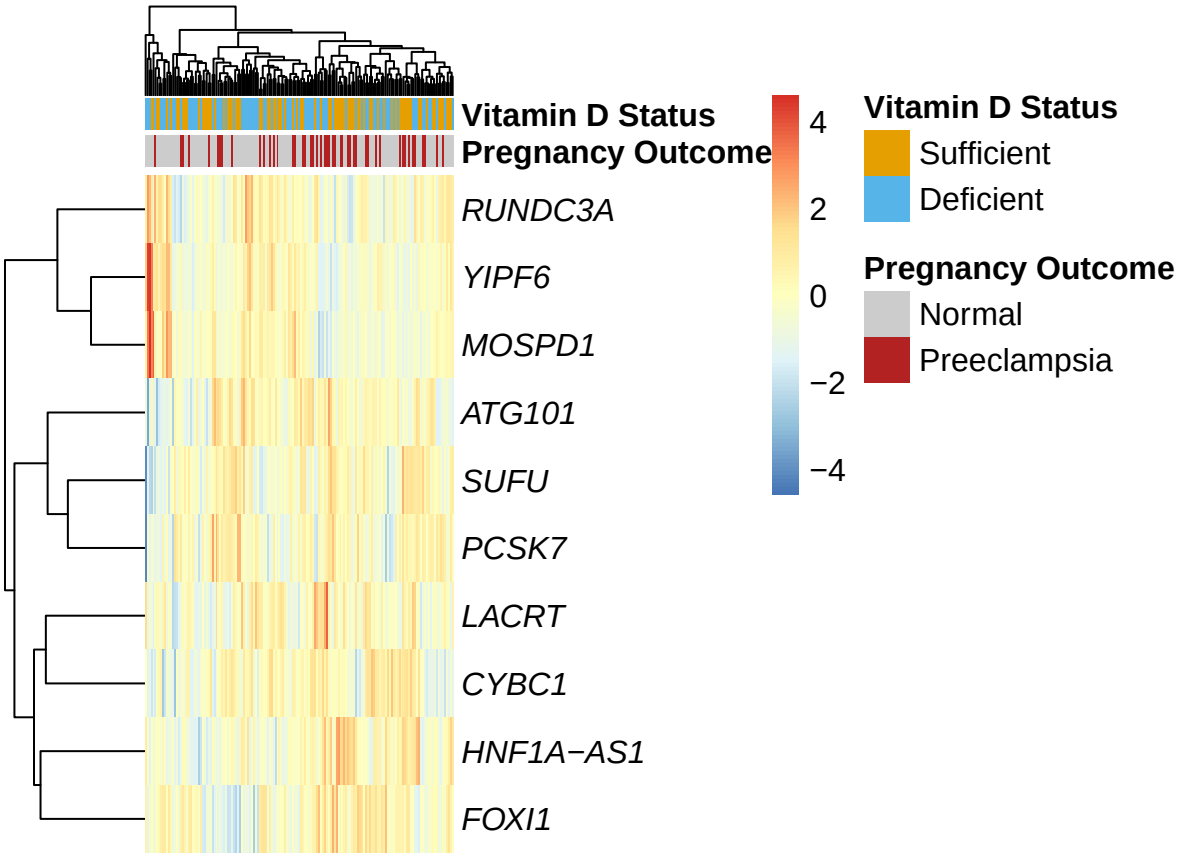
	Normal (n = 110)	Preeclampsia (n = 47)
Maternal age (years), mean (SD)	26.63 (5.08)	25.78 (4.95)
Maternal BMI (kg/m^2), median [IQR]	26.54 [22.35, 30.97]	31.24 [24.62, 36.41]
Gestational age at delivery (weeks), median [IQR]	39.29 [38.57, 40]	39.14 [36.93, 40.07]
Vitamin D status, n (%)	Sufficient: 58 (52.7), Deficient: 52 (47.3)	Sufficient: 21 (44.7), Defici

Heatmap

```
# load pheatmap
library(pheatmap)
# top 10 genes from preeclampsia mean difference ranking
top10 <- gene_rank$GENE[1:10]
# keep the 10 and use gene names as row names
heatData <- expr_named[expr_named$GENE %in% top10, ]
row.names(heatData) <- heatData$GENE
# matrix of genes (rows) by samples (colums), in same sample order as covar
heatMat <- as.matrix(heatData[, covar$geoID])
```

```
# Tracking bars
# data frame of 2 catVars with sample ID as row name
annot <- data.frame(row.names = covar$geoID, covar$preg_preeclampsia, covar$vitD_status)
colnames(annot) <- c('Pregnancy Outcome', 'Vitamin D Status')
# colors
annColors <- list('Pregnancy Outcome' = c(Normal = 'grey80', Preeclampsia = 'firebrick'),
                  'Vitamin D Status' = c(Sufficient = '#E69F00', Deficient = '#56B4E9'))
```

```
# Draw Heatmap
# adjust width after seeing result
fig.width = 25
# italicize gene names for row labels
geneNames <- as.expression(lapply(row.names(heatMat), function(g) bquote(italic(.(g)))))
# drat teh clustered heatmap with tracking bars
myHeat <- pheatmap(heatMat,
                   scale = 'row',
                   treeheight_col = 30,
                   clustering_distance_rows = 'euclidean',
                   clustering_distance_cols = 'euclidean',
                   clustering_method = 'complete',
                   annotation_col = annot,
                   annotation_colors = annColors,
                   labels_row = geneNames,
                   show_colnames = FALSE,
                   fontsize = 12)
```




```
# save heatmpa as PNG for overleaf
ggsave('heatmap.png', myHeat$gtable, width = 25, height = 6)
```

Beeswarm of the Gene Screen

```
library(ggbeeswarm)
# label each gene a p<0.05 or not
gene_rank$Significance <- factor(ifelse(gene_rank$p_value < 0.05, 'p < 0.05', 'p >=
↪  0.05'), levels = c('p >= 0.05', 'p < 0.05'))
gene_rank <- gene_rank[order(gene_rank$Significance), ]

# beeswarm of all 500 genes' mean differences, colored by p-value
myBees <- ggplot(gene_rank, aes(x = '', y = mean_diff, fill = Significance, color =
↪  Significance)) +
  geom_quasirandom(shape = 21, size = 3, stroke = 0.8) +
  geom_hline(yintercept = 0, linetype = 'dashed') +
  scale_fill_manual(values = c('p >= 0.05' = '#9ECAE1', 'p < 0.05' = 'firebrick')) +
  scale_color_manual(values = c('p >= 0.05' = '#6BAED6', 'p < 0.05' = 'black')) +
  coord_flip(ylim = range(gene_rank$mean_diff) * 1.1) +
  labs(x = NULL, y = 'Mean Difference in Expression (Preeclampsia - Normal, log2)', color
  ↪  = NULL, fill = NULL) +
  theme_classic(base_size = 16) +
  theme(legend.position = 'bottom', axis.ticks.y = element_blank())
```

Save beeswarm for Overleaf

```
# myBees
ggsave('beeswarm.png', myBees, width = 10, height = 6)
# print R and package versions for the Methods section
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ggbeeswarm_0.7.3 pheatmap_1.0.13  knitr_1.52       lubridate_1.9.5
```

```
##  [5] forcats_1.0.1    stringr_1.6.0    dplyr_1.2.1      purrr_1.2.2
##  [9] readr_2.2.0      tidyr_1.3.2      tibble_3.3.1     ggplot2_4.0.3
## [13] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] utf8_1.2.6         generics_0.1.4     stringi_1.8.9      lattice_0.22-9
##  [5] hms_1.1.4          digest_0.6.39      magrittr_2.0.5     evaluate_1.0.5
##  [9] grid_4.6.1         timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0
## [13] Matrix_1.7-5       mgcv_1.9-4         scales_1.4.0       textshaping_1.0.5
## [17] cli_3.6.6          rlang_1.3.0        splines_4.6.1      withr_3.0.3
## [21] yaml_2.3.12        otel_0.2.0         tools_4.6.1        tzdb_0.5.0
## [25] vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5    vipor_0.4.7
## [29] ragg_1.5.2         beeswarm_0.4.0     pkgconfig_2.0.3    pillar_1.11.1
## [33] gtable_0.3.6       glue_1.8.1         systemfonts_1.3.2  xfun_0.60
## [37] tidyselect_1.2.1   rstudioapi_0.19.0  farver_2.1.2       htmltools_0.5.9
## [41] nlme_3.1-169       rmarkdown_2.32     labeling_0.4.3     compiler_4.6.1
## [45] S7_0.2.2
```